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Subdued surface expression of deep cave collapse

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Cave systems can be difficult to recognise where karst landforms are subdued or overprinted with other landscape processes. Subtle linear depressions on carbonate platforms are commonly interpreted as fluvial or tectonic landforms, but may also mark collapse above underlying cave systems. We test this on the Nullarbor Plain, southern Australia, a stable >200,000 km² carbonate platform exposed for >14 Myr. Integrating morphometrics, electrical resistivity tomography, passive seismic profiling, cave records, cliff exposures and sedimentology, we show that kilometre-scale, shallow, sediment-filled trenches are surface expressions of vertically propagated collapse and sagging above deep cave conduits. These depressions lack integrated drainage, downstream sediment transport and sedimentological evidence of channelised flow; instead, they overlie vertically extensive low-resistivity zones, align with mapped caves and collapse dolines, and project to cliff-face exposed stratal deformation. The Nullarbor trenches provide diagnostic criteria for recognising subdued cave-collapse systems on Earth and an analogue for interpreting linear depressions on other rocky planetary surfaces.

Caverniferous landscapes most commonly develop through dissolution of soluble rocks (commonly carbonates and evaporites); known as karst terrains, these comprise ca. 15% of Earth's ice-free terrestrial surface^{1,2}. Potentially analogous caverniferous landscapes have been identified on other planets, including thousands of candidate cave entrances^{3–5}. On Earth, caves and cave deposits can persist as relict elements in the landscape, with passage morphology and sediment fills recording hydrological regimes that no longer exist at these sites^{2,6}. The relative stability of cave environments and the protection they provide from surface disturbance promote the preservation of palaeoenvironmental archives, archaeological remains and organic materials^{7–10}. This preservation potential is also relevant for planetary caves, which may provide accessible subsurface environments where organic matter or biosignatures, should they ever have existed, could be protected from surface radiation, temperature extremes and weathering^{9,11}. Recognition of caves is therefore vital for reconstructing past climates, environments and the history of life on Earth, as well as for assessing astrobiological potential on other worlds.

Many low-gradient cave passages in karst landscapes develop under phreatic or epiphreatic conditions at or below the watertable and are generally hidden from view¹²; even after they are drained they can be difficult to locate. Therefore, where such cave passages influence the ground surface above them, their surface expression provides an important constraint on their spatial position. The most common surface geomorphological features linked to caves form when the mechanically unstable roof of an underground cavity collapses; this may propagate the cavity upwards until the surface is breached^{6,13,14}. The clearest examples are collapse dolines, which are closed and relatively circular depressions¹⁵. Collapse of larger segments of cave passages can form aligned dolines or linear valley-like depressions that can be readily linked to subsurface cave systems², e.g. Peruaçu Gorge in Minas Gerais, Brazil¹⁶, Stockyard Gully in Western Australia¹⁷, and a section of the Grand Canyon in Arizona, USA¹⁸. Additional surface expressions of karst systems are unroofed caves^{19–21}; these form by gradual chemical denudation (surface lowering), preserving the original conduit morphology and often

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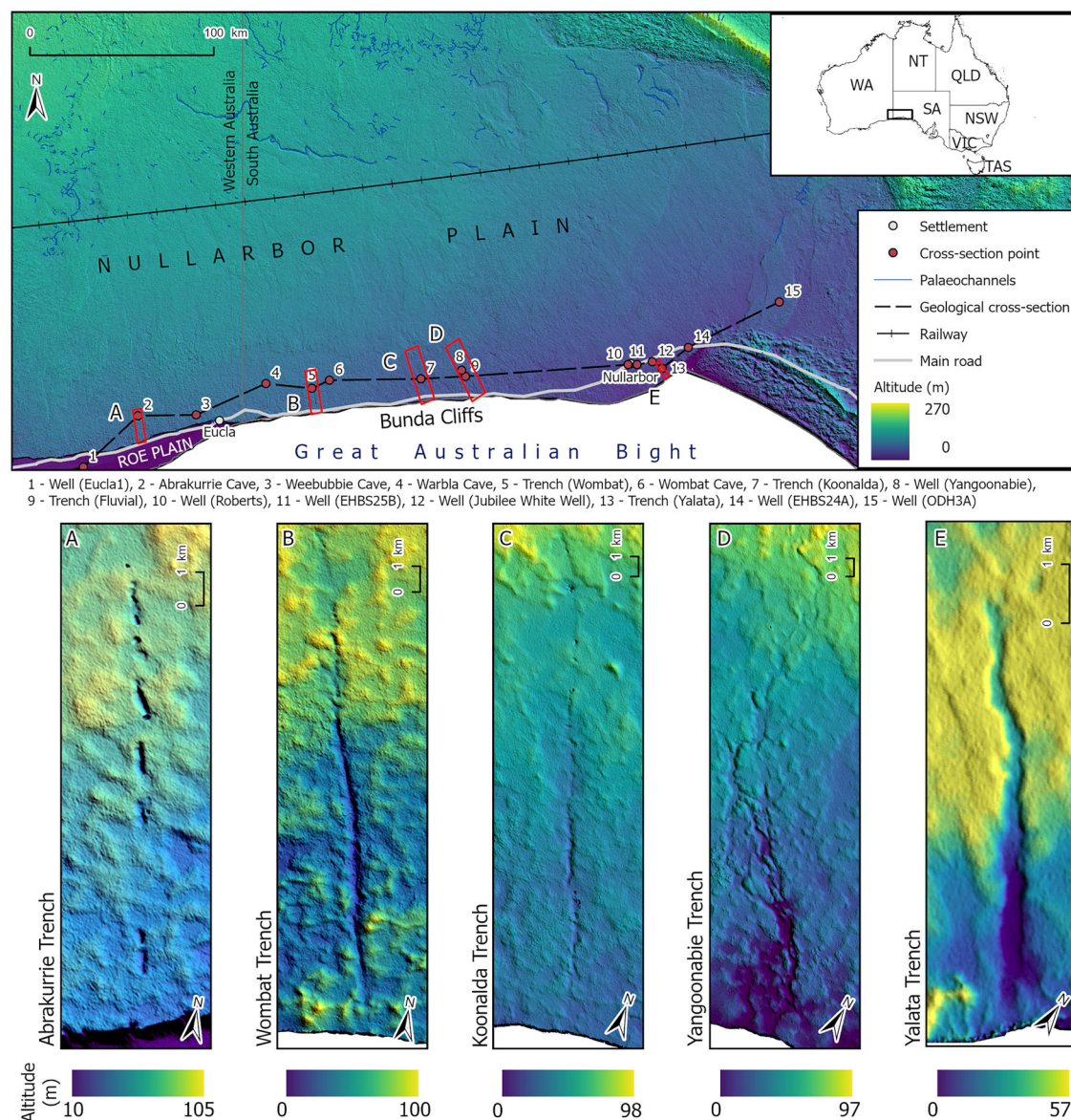


Fig. 1 | Locality map of shallow trenches on the Nullarbor Plain. Note that the digital elevation models are exaggerated for clearer presentation of the trenches. The upper panel shows the regional digital elevation model (DEM) and locations of key features and sites, including points defining the geological cross-section shown in

Fig. 4, while panels A–E show enlarged DEM views of the Abrakurrie, Wombat, Koonalda, Yangoonable and Yalata trenches, respectively. Basemap sources: German Aerospace Centre TanDEM-X; The Shuttle Radar Topography Mission (SRTM) digital elevation dataset.

retaining flowstone and stratified clastic sediment infills as clear evidence of their speleogenetic origin.

On low-relief, tectonically stable carbonate platforms in arid climate regions, the surface expression of cave collapse is generally subdued, because gradual surface lowering (due to slow dissolution rates), calcretisation and aeolian infill obscure collapse morphologies as they develop. As a result, cave collapses are difficult to recognise, even though deep karst voids can persist for tens of millions of years. Collapsed cave passages may be present only as subtle linear depressions that are easily overlooked or misinterpreted as fluvial, tectonic or even aeolian-related landforms. In more tectonically and/or hydrologically active settings, collapsed cave features may be equally subtle and can be overlooked entirely due to the dominance of other, more pronounced surface structures. As a result, cave collapse systems, potentially globally widespread, are often undetected because their surface signature is difficult to identify.

Here we resolve this problem by demonstrating diagnostic criteria for recognising low relief cave-collapse systems on the arid Nullarbor Plain of

southern Australia, which is the largest onshore exposure of Cenozoic carbonate marine sediments anywhere in the world²². The Nullarbor Plain is tectonically stable and compositionally homogeneous, and despite being subaerially exposed for over 14 million years (Fig. 1), represents an exceptional archive of subtle depositional, structural and environmental structures^{23–25}. By integrating morphometrics, electrical resistivity tomography, passive seismic profiling, stratigraphy and sediment analysis, we demonstrate that shallow, sediment-filled linear depressions on the Nullarbor Plain represent long-term upward propagation of collapse from cave systems to the surface, rather than shallow fluvial valleys, which they resemble in planform geometry.

The Nullarbor Plain surface depressions have evolved under prolonged aridity, allowing collapse morphologies to persist without significant fluvial reworking. Thus, they offer a template for recognising collapse trench formation in dry settings on other planets. This may apply both to potential karst settings, e.g. on Titan and Mars^{26,27}, as well as to volcanic terrains, e.g. on Mars, where elongate troughs have been interpreted as evidence of

surface runoff, but could equally reflect subsurface void development, particularly where they are associated with atypical collapse pit craters^{1,28,29}.

Regional setting

The Nullarbor Plain is a low-relief carbonate platform covering more than 200,000 km² of the southern margin of Australia. It is composed of a gently southward, seaward-dipping succession of Cenozoic limestones separated by subtle disconformities: Middle-Upper Eocene Wilson Bluff Limestone^{30–32}, Upper Oligocene-Lower Miocene Abrakurrie Limestone^{30–32}, and Middle Miocene Nullarbor Limestone^{30–34}. Since the sea regressed in the Middle Miocene and exposed the platform to weathering, the region has experienced long-term landscape stability with minimal tectonic deformation, no glaciation and negligible deposition of sediment, allowing geomorphic records of past environments to persist. These include longitudinal dunes etched into the limestone²⁵ and topographically elevated primary depositional bioherms preserved in the Nullarbor Limestone²³. The present surface relief is extremely subdued, surface drainage is absent or ephemeral, and surface karstification is weakly expressed. The southern margin of the Nullarbor Plain is an extensive cliff-line, the Great Southern Scarp, which is among the longest uninterrupted escarpments in the world³⁵.

Sub-horizontal cave systems are a characteristic feature of the Nullarbor Plain³⁶. Caves on the Nullarbor Plain are typically categorised as “shallow” or “deep”^{37–39}, extending <30 m and 50–150 m below the surface, respectively⁴⁰. Shallow cave passages are confined to the Nullarbor Limestone and range up to 200 m in length³⁷. They are often characterised by sub-horizontal passages with abundant deposits of dark brown calcite, as well as halite and gypsum⁴⁰. These shallow caves are associated with pocket valleys in the Hampton and Wylie Scarps⁴¹, and their genesis is associated with mixing corrosion during the Late Miocene - Pliocene sea-level highstand⁴².

Deep caves occur predominantly within ~60 km of the southern coast. Most of them are accessed through collapse dolines and contain passages extending hundreds of metres and reaching tens of metres in width and height⁴⁰. Besides the significant size of their passages and chambers, the deep caves are characterised by extensive collapse, general absence of calcite deposits, and continuation below the present water table. Halite and gypsum speleothems may be present⁴³. The formation of deep caves is unrelated to the formation of shallow caves. These caves developed predominantly within the older Wilson Bluff Limestone as extensive trunk conduits, probably during periods of higher effective recharge in the Oligocene to Early Miocene, when the sea withdrew from the carbonate platform and allowed cave development⁴⁰. The orientation of known cave passages suggests the caves formed by coastward-flowing groundwater, with conduit enlargement increasing southwards toward the palaeocoast. Following final regression and uplift in the Late Miocene–Early Pliocene, lowering of the watertable drained the caves and removed buoyancy, initiating progressive roof failure, so that the cave passages are now characterised by extensive collapse^{38,44}.

Results

Along the eastern edge of the Nullarbor Plain, close to the Great Southern Scarp, five trenches (elongated linear to gently curvilinear depressions) are developed, oriented generally NNW-SSE, ranging from several kilometres to more than 20 km in length and generally 100–500 m wide, yet typically less than 9 m deep (Table 1; Fig. 1). These trenches have sediment-filled floors and subdued margins. They are not integrated into any surface drainage network and show no evidence of downstream sediment transport. Sediment within the trenches becomes progressively finer-grained with depth. Sand dominates the shallow upper layer, whereas deeper layers show progressively higher proportions of silt and clay. The sediment is composed of calcite, quartz and clay (illite and kaolinite) with traces of feldspars, dolomite, aragonite, halite, anatase and gypsum (see the Supplementary material for the complete analytical data).

The westernmost Abrakurrie Trench (Figs. 1A and 2A), occurs as a discontinuous alignment of collapse dolines and shallow depressions. It is spatially associated with known cave systems aligned along the trench and

Table 1 | Morphological characteristics of studied trenches from west to east

Trench system	Total length (km)	Orientation (°)	Width (m)	Surface relief (m)	Continuity	Relationship to collapse dolines and caves
Abrakurrie Trench	~12	151–169 (NNW-SSE dominant)	Up to ~300	Up to ~9; dolines to 85 m	Entirely fragmented	Numerous major collapse dolines (Chowilla Landslip, Abrakurrie, Kutowalla, Winbirra); direct cave connections; local blind-valley behaviour
Wombat Trench	~20	161–170 (NNW-SSE)	Up to ~350	1–2 (N); up to ~9 (S)	Discontinuous north; continuous south	Small collapse dolines and rock pockets along trench; no direct cave association; nearest cave ~9 km east
Koonalda Trench	~22	155–166 (NNW-SSE)	Up to ~300	up to ~9	Discontinuous north; continuous south	Multiple collapse dolines; direct connection to Koonalda Cave (N-4); Clay Dam Cave exposes sagging strata and infill stratigraphy
Yangoonable Trench	~28	109–180 (NW-SE to N-S)	Up to ~2000 (branched zone), up to ~350 (single trenches)	up to ~5	Highly fragmented north, then branched, locally subparallel, and more continuous south	Collapse dolines occur within ~3–4 km but not within trench; margins marked by rockholes and blowholes
Yalata Trench	~7	~150 (NNW-SSE)	Up to ~350	up to ~5	Continuous	No caves or dolines within trench; isolated dolines ~3–4 km away; nearest cave White Wells Cave (N-14; ~4 km NW)

extending beneath sections where no trench occurs (e.g., Abrakurrie Cave (N-3); Supplementary Fig. 5B), so there is a clear geomorphic link between the surficial trench and subsurface voids. Likewise, Koonalda Cave (N-4; Supplementary Fig. 9) forms a ~1 km long subterranean continuation

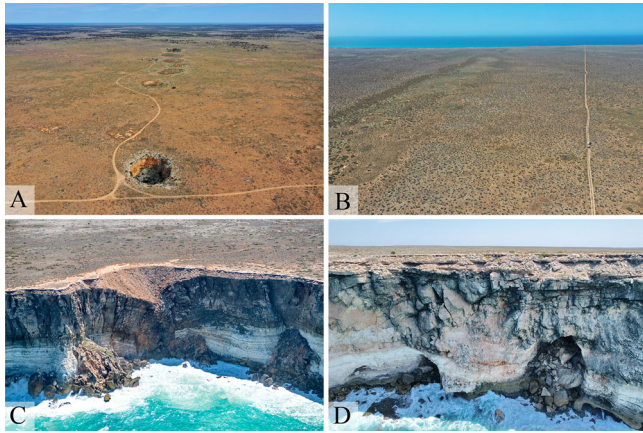


Fig. 2 | Surface and cliff expressions associated with trench systems. Unmanned aerial vehicle photographs of collapse dolines within the westernmost Abrakurrie Trench (A) and easternmost Yalata Trench (B), and cliff-exposed deformation at the southern projected extension of the Yangoonabie Trench (C) and Yalata Trench (D); note the sagging strata. C Photo courtesy of Ian Oswald-Jacobs.

northward of Koonalda Trench. Further east, the trench systems become wider with more laterally persistent linear depressions, but contain fewer, if any, collapse dolines or cave entrances (Figs. 1B–E and 2B).

Cocklebidy Cave (N-48), which is located ~250 km west of the Abrakurrie Trench and therefore not part of the mapped trench systems, provides an important regional analogue for the subsurface cavities inferred beneath the trenches^{40,45}. This cave is ~6 km long, approximately linear NNE–SSW, lies ~100 m below the surface and consists of 20–50 m wide collapse domes and passages, many of which are water-filled. Its geometry and scale demonstrate that long, linear cave systems exist beneath the Nullarbor surface without forming a continuous surface trench. This provides a useful subsurface analogue for the mapped trenches and supports the possibility that linear surface depressions may reflect upward propagation from similar subsurface cavities.

The subsurface data provide further evidence for this relationship. Electrical resistivity tomography (ERT) profiles revealed that the shallow trenches overlie low-resistivity zones that extend beyond the depth limit of our investigation (~30–70 m) and are surrounded by higher resistivity coherent limestone (Fig. 3B; see Supplementary material for all ERT profiles). In all trenches, there was a wide, shallow, low-resistivity zone in the upper 5–10 m beneath the flat trench floors (Fig. 3A, B), that is infilled with sediment. The near-surface trench widths exceed the maximal widths of known cave passages on the Nullarbor, most likely because of widening by mechanical erosion and lateral dissolution⁴⁶ along the trench margins.

Beneath the laterally expanded upper low-resistivity zone, ERT profiles commonly display a distinct ledge a few metres thick, sometimes developed symmetrically along the trench margins (Fig. 3A, B). This is interpreted as a

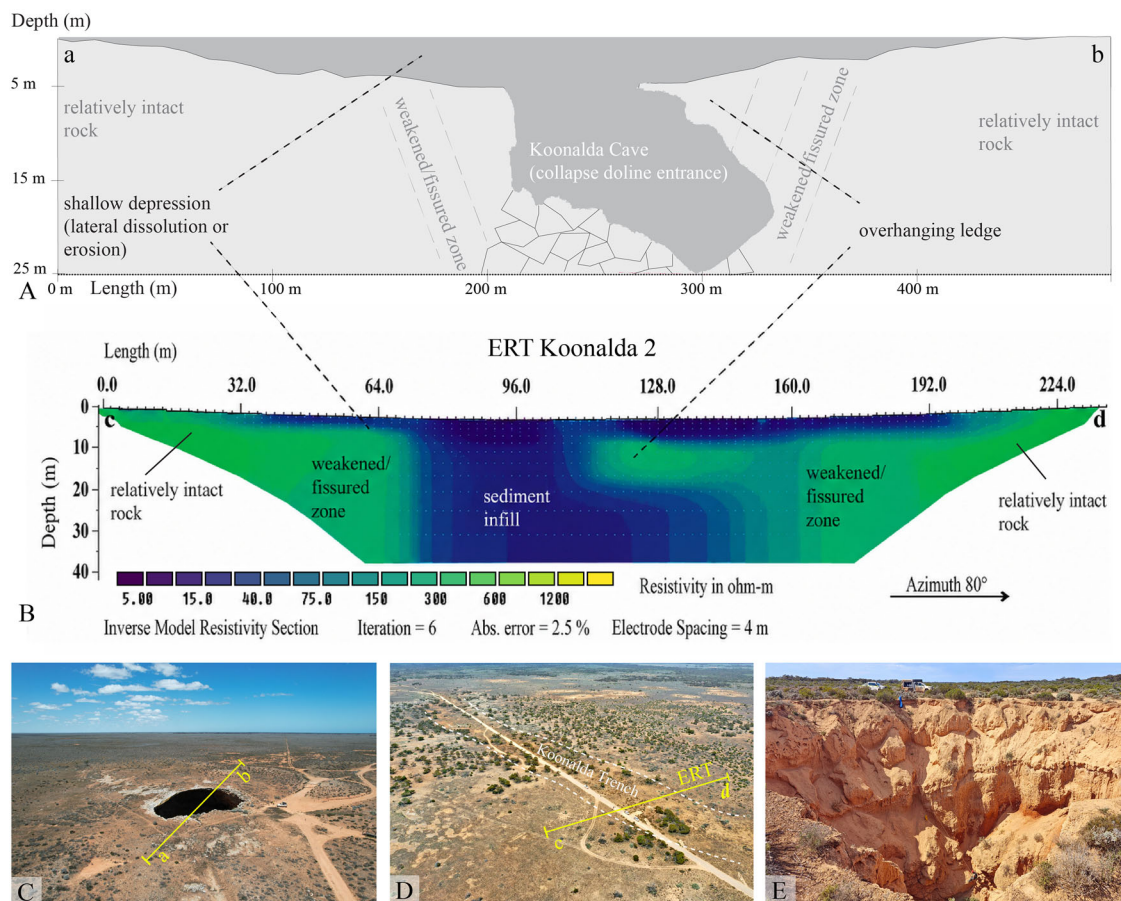


Fig. 3 | Surface and subsurface structure of the Koonalda Trench and associated caves. Cross-section of the Koonalda Cave collapse doline entrance constructed from unmanned aerial vehicle photogrammetry (A; see C for aerial view of the Koonalda Cave entrance), and ERT profile of the Koonalda Trench (B; see

D for aerial view of the Koonalda Trench; see Supplementary Fig. 9 for location). Clay Dam Cave entrance (E) is situated within the Koonalda Trench, exposing the thickness and characteristics of the sediment, identified as low resistivity region within the ERT profile.

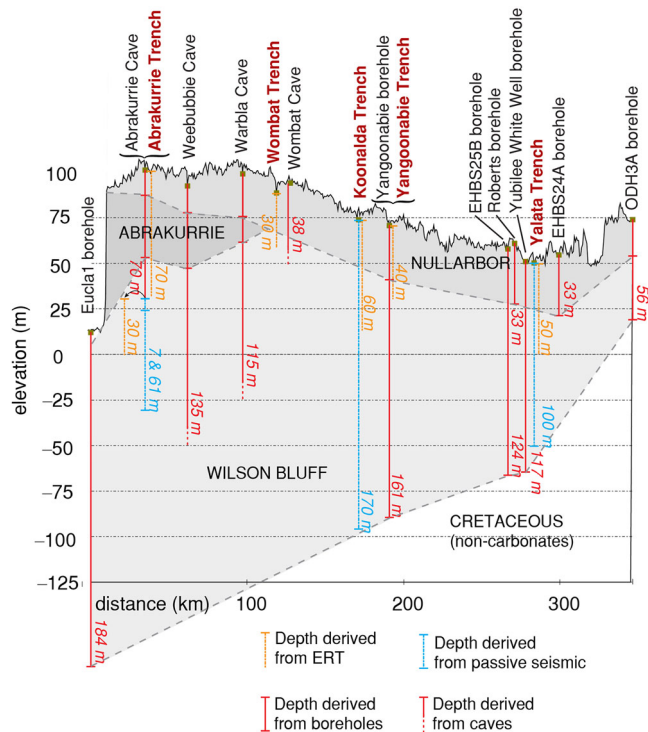


Fig. 4 | Geological cross-section (west to east) over the region where trenches occur. Depth constraints of stratigraphic boundaries derived from ERT (sediment) and caves should be considered as minimum depths. See Fig. 1 for cross-section location.

mechanically more resistant limestone layer; its depth corresponds closely to the stratigraphic boundary between the Nullarbor Limestone and the underlying Wilson Bluff Limestone, as shown by nearby boreholes.

Passive seismic measurements perpendicular across the Koonalda Trench show that the velocity contrast interpreted as the boundary between the Wilson Bluff Limestone and underlying Cretaceous siliciclastics (Fig. 4) occurs slightly deeper beneath the trench axis than beneath its margins (Supplementary Fig. 19). This is interpreted to be a result of reduced seismic velocities beneath the trench axis, indicating a zone of relatively less consolidated or more fractured material, consistent with the ERT results.

Direct confirmation of the postulated subsurface architecture beneath the trenches is provided by Clay Dam Cave (N-16), a collapse doline within Koonalda Trench. Its ~20 m deep collapse exposes sagging strata and sandy loam sediment infill (Fig. 3E) that match the geometry and depth of the low-resistivity zone imaged by ERT across this trench (Fig. 3B), offering a rare opportunity to directly observe the internal structure inferred beneath the other trenches.

Additional independent verification of the relationship of trenches to cave systems and zones of fractured rock is provided by exposures where the trenches are projected to intersect the coastal cliffs. In these locations, subsurface collapse zones, commonly over 100 m wide, show sagging strata (Fig. 2C), vertical brecciation zones, collapse arches and caves (Fig. 2D). The cliff-top surface is geomorphologically subdued and shows little evidence of the collapse below, due to intact cave roof strata (incomplete surface propagation), reinforcement by calcrete development and, in places, an aeolian sediment cover, illustrating the decoupling between surface relief and subsurface structures. Together, the spatial association of trenches with mapped cave passages, vertically extensive geophysical anomalies, exposed sagging and sediment infill at Clay Dam Cave, and cliff-face collapse structures provide mutually independent lines of evidence that the surface trenches are linked to vertically continuous zones of collapse above subsurface voids.

There is no evidence that the trenches are of an aeolian or fluvial origin. They cannot be relict interdune swales (which are preserved elsewhere on the Nullarbor Plain as shallow linear depressions with negligible sediment²⁵)

because they are not underlain by solid limestone and contain a substantial sediment thickness.

A primary fluvial origin is inconsistent with the absence of an integrated surface-drainage network in the trench area; the trenches begin and end abruptly, lack tributary catchments or upstream source areas beyond the karst, and cannot be traced into hinterland drainage systems. Relict fluvial palaeochannels on the inland margins of the Nullarbor Plain^{24,47} have low-energy meandering planforms unless structurally confined, and none extend to the southern margin of the plain where the trenches occur, because surface flow was progressively lost by infiltration and evaporation³⁸. Nevertheless, the trenches show some evidence of later hydrological modification. They contain a basal layer of silt and clay that was probably washed in from the sides, where it was originally derived by weathering of the limestone, which has a small silt/clay component. Depressions on the Nullarbor currently receive sediment during rare intense rainfall events⁴⁸, and the wetter Pliocene climate on the Nullarbor⁴⁹ would have enhanced sediment wash-in.

The trenches are not solely tectonic in origin, as there is no surface vertical displacement, as is present along the numerous faults that have been documented across the southern part of the Nullarbor Plain²⁴. However, the trenches may be structurally controlled features, as their broadly north-south orientation is parallel to the documented faults.

Instead, the depressions display a pronounced rock-mass deficit that requires subsurface removal of material, consistent with dissolutional formation of cave voids followed by roof collapse that has progressively propagated upward to the surface¹⁵. Discontinuous trenches, e.g. Abrakurrie Trench (Fig. 1 & Supplementary Fig. 5B), contain sections where upward collapse has not yet reached the surface, as shown by the presence of collapse dolines connecting to caves beneath these sections.

Chronological framework of trench development

The five trenches in this study show a west–east transition from cave-connected collapse dolines to continuous sediment-filled trenches (Fig. 1), corresponding closely to a decrease in the thickness of the carbonate stratigraphy from ~250 m in the west near Abrakurrie Trench to ~110 m in the vicinity of Yalata Trench (Fig. 4). The original cave passages probably formed at or towards the base of the carbonate sequence. The eastern trench systems near the basin margin appear to represent a more advanced stage of development, where the cave collapses propagated upwards through a smaller thickness of limestone in a shorter time. The discontinuous collapse features in the west are interpreted to reflect a less advanced stage in which collapse has not yet fully propagated to the surface through the thicker limestone.

The systematic west–east variation in trench morphology therefore provides a chronological framework for the long-term evolution of cave-collapse systems under stable conditions, preserving a continuum from readily identifiable collapse dolines to shallow, subdued and barely perceptible surface footprints.

This developmental sequence on the Nullarbor Plain is not purely relict, as shown by a mid-Pleistocene (~460 ka) U-Th date on a calcrete layer 3 m below the present-day surface calcrete at Clay Dam Cave in Koonalda Trench. The presence of older calcrete layers beneath the dated layer demonstrates that although present-day karst processes operate at low rates, they are sufficient to maintain a net rock-mass deficit, so collapse, subsidence and infilling have occurred repeatedly and probably continue today, as indicated by fresh rock collapses from cave ceilings and collapse doline walls⁵⁰.

Recognition of collapsed cave systems

The Nullarbor Plain provides a global reference example for recognising collapsed cave systems whose surface expression is visible but subdued, i.e. low-relief, sediment-filled and lacking visible steep collapse walls or canyon-like morphology that make many cave-collapse valleys readily identifiable. Just as palaeo-fluvial channels are routinely identified as geomorphic records of former drainage systems, so laterally extensive, shallow trenches

in karst terrains should be considered in case they represent surface expressions of deep conduit collapse. The diagnostic criteria established here (shallow linear depressions underlain by vertically extensive low-resistivity zones, collapse-related subsurface voids and mapped cave passages) offer a transferable framework for recognising relict collapse systems where the cave-collapse origin is not evident from surface morphology alone. This framework can identify subsurface void architecture, so aligned depressions may indicate continuity of linear passages, whereas more reticulate or irregular depression patterns may point to more complex maze-like cave geometries. Recognising these features bridges the gap between subsurface (palaeo)karst architecture and surface geomorphology, and expands the range of landscape forms that can be interpreted as products of cave collapse.

The subdued but persistent collapse-generated trenches on the Nullarbor Plain provide an analogue for recognising hidden, laterally extensive subsurface void systems from subtle surface depressions in arid, low-relief terrains. In planetary settings, linear depressions and collapse alignments may therefore help contextualise candidate cave entrances^{5,51} within broader subsurface passage systems. The transferable element of our study is therefore diagnostic; collapse alignment, spatial clustering of morphologically similar depressions and the absence of an integrated surface-drainage network can be integrated with available topographic, geological history and where available, geophysical constraints to prioritise candidate cave-bearing regions and infer the presence of possible lateral passages^{4,5,51}. For example, comparable associations of dry valley networks and collapse or doline-like morphologies have been reported from Mars^{4,52,53}. The Nullarbor Plain therefore provides a terrestrial end-member for understanding how subsurface void development, collapse and limited surface drainage may interact on rocky planetary surfaces where climatic signals are weak but geomorphic preservation is likely to be high.

Methods

Australian Speleological Federation (ASF) Ozkarst database of caves and karst features was used to locate features and obtain their basic characteristics. Each cave on the Nullarbor Plain has been assigned a unique identification number with a code (e.g., N-01). These identifiers are retained in this study to maintain consistency with the established cave registry.

Drillhole stratigraphical and hydrological data were obtained from WAPIMS maintained by the Department of Energy, Mines, Industry Regulation and Safety, Government of Western Australia, GeoVIEW.WA (<https://geoview.dmp.wa.gov.au>) maintained by the Department of Mines, Petroleum and Exploration, Geological Survey of Western Australia, and from the WaterConnect webpage (<https://www.waterconnect.sa.gov.au>), maintained by the Department for Environment and Water, Government of South Australia.

The surface topography was studied using Global Mapper (Blue Marble Geographics) software applied to TanDEM-X digital elevation model (DEM; 0.4 arc second and 1.0 arc second), obtained from the German Aerospace Centre⁵⁴. To align the vertical datum of the DEM, originally based on the ellipsoid, with a geoid-based reference system, we employed ESRI ArcGIS tools for the conversion process. The geoid raster (downloaded from http://earth-info.nga.mil/GandG/wgs84/gravitymod/egm2008/egm08_gis.html) was clipped and resampled to match the tile from the TanDEM-X data. The TanDEM-X DEM was then converted by applying the equation 'DEM minus Geoid'. The visualisation of the DEM data was additionally enhanced by the Relief Visualization Toolbox (RVT) created by ZRC SAZU^{55,56}, using prismatic openness for blend combination.

Satellite images were also used and accessed through the Global Mapper World Imagery online data source, attributed to Esri, Maxar, Earthstar Geographics, CNES/Airbus DS, USDA FSA, USGS, AeroGRID, IGN, IGP, and the GIS User Community.

Electrical resistivity tomography

Electrical resistivity tomography (ERT) is a relatively fast, low impact geophysical method to portray differences of electrical subsurface

properties^{57–59}. Physical-chemical sediment properties such as mineralogy, texture, soil moisture levels, salinity or temperature all have some influence on the electrical properties of subsurface material and thus respond to electrical fields⁶⁰. 2-D ERT measurement lines provide the opportunity to interpret measured differences in a geomorphic-sedimentological way.

We used the multielectrode system '4pointlight hp' from Lippmann Geophysical Instruments⁶¹ with a maximum of 100 steel electrodes with varying spacings between 2 and 5 m. A roll-along technique was used to extend ERT lines where necessary. Early tests showed the Wenner array as an effective electrode configuration with a stable and good signal to noise ratio. Wenner describes a 4-electrode array with active current electrodes on the outside (electrode 1 and 4) and the two inner electrodes (electrode 2 and 3) measuring the voltage drop. The apparent resistivity (ρ_a), measured in Ωm (denoting Ohm's law), is calculated by Loke, Chambers⁶²:

$$\rho_a = \frac{2\pi k \Delta V}{I} \quad (1)$$

where ΔV is the voltage drop (V), k is the spacing between electrodes and I is the measured current (A).

Pseudo-sections detailing the geometry of different electrode configurations and spacings were planned and executed using Geotest software⁶¹ with a measurement frequency of 4.16 Hz and electric currents between 0.1 and 100 mA. Each point was measured three times and the average was stored if the value was below a 3.0% difference threshold. If the value was larger than 3.0%, the measurements were repeated eight times and averaged.

ERT has been tested on the Nullarbor Plain before in search of open cave passages, but with limited success due to dry and well insulated ground⁶³. However, we collected field data in October 2022 and October 2024. In both years, rainfall during or before our measurements allowed low contact resistance ensuring robust field data.

Data inversion using RES2DINV software 3.55.18⁶⁴ generated 2D models of the specific resistivity by using linear least-squares optimisation inversion techniques. The chosen convergence limit of 3.0% was usually reached after five to six iterations with absolute error between 2.5 and 15% between measured and modelled apparent resistivities. Relative elevation level of each electrode was extracted from the drone survey and applied to the resistivity lines prior to the inversion.

Interpretation of ERT images was derived by separating resistivity zones by their physical values and shapes. These ERT-shapes were successfully identified in nearby outcrops and accessible cave systems corroborating the ERT-derived interpretation. Additionally, we also collected representative samples (sediments and rocks) to measure their electric properties under controlled conditions in the laboratory (see Supplementary material), which further consolidated our interpretation.

Passive seismic field set-up

In this study, we use the passive HVSR method⁶⁵ to record subsurface ambient noise caused by either human-induced forces such as noise from road or train traffic, or by natural forces such as wind, waves or earthquakes^{66,67}. Passive seismic was mainly applied to depths below ERT's depth of exploration at >30–40 m depth and to estimate depth to bedrock where ERT was inconclusive. High-frequency noise in the signal (often >25 Hz) is mostly attributed to windy conditions during the measurement. Strong winds are frequent in the Nullarbor and were identified as an issue during previous attempts using passive seismics⁶³.

We used a Tromino ENG 3G seismometer (MOHO, Venice Italy) with a sampling rate of 128 Hz to collect HVSR data along three axes (Z component up-down, Y component east-west, X component north-south). We oriented the instrument to north and ensured tight coupling to the ground. Measurements were taken for 20 min and windy days were avoided to ensure better data quality and less noise.

Grilla software (MOHO) was used to analyse the field data. Noisy sections of the 20 min measurement window were removed for the subsequent analysis which were restricted to 0.1–64 Hz. We used the 0.1–20 Hz

range for stratigraphic interpretation. Thickness of depth to bedrock (or major refraction surface) was calculated as following:

$$f_n = \left(\frac{V_s}{4h} \right) \quad (2)$$

where f_n is the resonance frequency associated with a seismic boundary (Hz), V_s is the shear-wave velocity and h is the thickness of weathering profile (m). We estimated shear wave velocities by applying data from previous studies (see Weller et al.⁵⁹). Additionally, simple refraction seismic line at the bottom of the Abrakurrie Cave was processed as dispersion curves and yielded shear wave velocities of 340–350 m/s. Subsequently, we developed 1-D models for several locations.

Drone photogrammetry

Drone photography was carried out using a DJI Mavic 2 Pro and a DJI AIR2S drones. The aerial imagery was obtained in the field to provide higher resolution data to complement regional satellite images, digital elevation models and ERT results. This was done to determine the accurate digital elevation model for ERT lines, to document morphology of collapse caves, and to document the walls of the Nullarbor (Bunda) Cliffs. At each ERT site several hundred photographs were taken at an angle of 90° from heights of 20–30 m with the UAV's factory-installed camera. Similarly, at each point on the cliffs that was correlated to the closest point of the theoretical outflow of the linear depressions, the cliffs were recorded 1 km in each lateral direction. The recordings allowed us to produce images with a resolution up to a few cm.

The acquired imagery was then processed in Agisoft Metashape Professional. First, images of inadequate quality were removed, i.e. those with a quality score of less than 0.5 as recommended by software developer. Then, photogrammetric methods (photogrammetry is a technique of evaluating forms, dimensions and positions of objects by analysing images of them⁶⁸) were used to produce a dense point cloud representing the surface. Further, on the basis of the point cloud, a digital elevation model and orthorectified image or orthophoto were produced (see also Peternel et al.⁶⁹ and Cignetti et al.⁷⁰ for the details of the methodology). The point recordings and the final products were prepared in the GDA2020 MGA projected coordinate system.

Geological analyses

Sediment samples from trenches (Abrakurrie Trench 31°38'49.5" S, 128°29'11.6" E; Wombat Trench 31°32'41.7" S, 129°20'29.3" E), including Abrakurrie Cave and Clay Dam Cave (cave locations can be obtained from Australian Speleological Federation Inc. (ASF)), were collected to analyse their grain size and mineralogy at the Faculty of Natural Sciences and Engineering, University of Ljubljana. Permit reference numbers are "RA-00001504; File A00031-24", "File 00514-1999; Case 2302956", and "E27407-1".

Sediment grain size analysis was performed using a FRITSCH Analysette 22-28 instrument combining a laser granulometer and dynamic image analysis. The size range of the measurements with the aforementioned instrument ranges between 10 nm and 2 mm. Sediment samples were first quartered using a stainless steel sample splitter to obtain a suitable amount for analysis (~1 g). Then, 3 ml of distilled water was added to the sample in a PVC container and homogenized for 5 min at a frequency of 25 Hz with a MillMix 20 shaker. The obtained data was used to classify sediments according to Blott and Pye⁷¹.

The mineralogy of samples was determined using X-ray diffraction (XRD) with an Empyrean III diffractometer (Malvern Panalytical) equipped with a MultiOptics module (iCore and dCore) and a 1Der detector on randomly oriented powder mounts. A spinning stage was used to ensure uniform sample exposure during measurements. The samples were finely ground to a particle size less than 30 µm to reduce the influence of the preferred orientation and placed in a back-loading

sample station for optimal surface alignment. Diffraction data were collected over a 2θ range of 3° to 70° with a step size of 0.004° and a counting time of 12 s per step. The instrument operated with a Cu X-ray tube at 45 kV and 40 mA, using a fixed divergence slit of 0.5° and a 14 mm incident beam mask. Diffraction patterns were processed using HighScore Plus version 5.3. Background determination, peak position and intensity were performed, followed by phase identification through comparison with patterns in the ICDD PDF-4 Minerals 2025 database. Quantitative phase analysis was conducted using the Rietveld refinement method, in which a polynomial function was fitted to the measured diffraction patterns. The refinement yielded weighted profile R-factors (R_w) ranging from 10.5 to 12.9 and goodness-of-fit (GOF) values between 1.2 and 2.9.

U/Th dating

Two calcareous sediments of the Clay Dam Cave of the Koonalda Trench were dated by the uranium-thorium (U/Th) method at the University of Queensland. Ages were corrected for non-radiogenic ²³⁰Th incorporated at the time of deposition. Full details of the method are provided in Hellstrom (2003, 2006)^{72,73}. Age errors are reported as 2σ uncertainties.

Data availability

The data generated and analysed in this study, including morphometric, geophysical (ERT and passive seismic), sedimentological and mineralogical data, are publicly available and stored in the Zenodo repository under <https://doi.org/10.5281/zenodo.20529581>⁷⁴. No restrictions apply to data access.

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Author contributions

Matej Lipar led the project, conducted field work, and led the manuscript writing. Matthias Leopold conducted field work and provided expertise on the electrical resistivity tomography and passive seismic profiling. John A. Webb contributed to the overall scientific direction, interpretation of results and writing and editing the manuscript. Rok Ciglić conducted field work and GIS analyses, and produced the maps. Jure Tičar conducted field work and supported the data analysis. Matija Zorn conducted field work. Uroš Stepišnik conducted field work and provided expertise on karst geomorphology. Matej Jelovčan conducted field work. Milo Barham contributed to the overall interpretation of results and writing and editing the manuscript. Matej Dolenec performed geological analysis and contributed to the interpretation of results. Primož Miklavc conducted field work, performed geological analysis and contributed to the interpretation of results. Tomislav Popit conducted field work, performed geological analysis and contributed to the interpretation of results. Andrej Šmuc conducted field work, performed geological analysis and contributed to the interpretation of results. Jian-xin Zhao performed the U-Th analysis and contributed to the interpretation of results. Mateja Ferik conducted field work and provided overall project oversight and scientific guidance throughout, including project conceptualisation and writing and editing the manuscript. All co-authors commented on the manuscript.

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Competing interests

The authors declare no competing interests.

Additional information

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